

BTH545 Workshop 4

Learning Outcomes

- Designing a user interface that uses widgets and layouts to build a user friendly GUI.

Design (60%)

You are to design an interface that will allow for the customization of toolbars in an application. The application has multiple tools which can be placed on any of three toolbars. There are a total of 30 different tools and each toolbar contain a maximum of 10 tools. Tools are identified by a textual name and a unique icon. The same tool can be placed on more than one toolbar but can only appear once on any one toolbar. In addition to adding tools to toolbars, you need to be able to remove them from toolbars. Your application should have a menu bar with commands for help, loading a saved toolbar configuration, saving a toolbar configuration, and exiting the application. You do not need to be able to load/save more than one toolbar configuration. Your application needs to satisfy the following constraints:

- **VISIBILITY:** You should be able to determine which tools are on which toolbar easily and determine which tools are not on any toolbar.
- **LEARNABILITY:** By using a simple metaphor, the application should be easy to learn and remember,
- **EASE OF USE:** There should be minimal steps involved in performing an operation.
- **CONSISTENCY:** The interface should operate in a similar way throughout. If you do an operation one way in a part of the interface, it should be done the same way in other parts of the interface.
- **ERROR PREVENTION:** Whenever possible, the interface should make it impossible to make mistakes.

This assignment is largely based on design. You should use whatever drawing tool you want and render an image of your interface for the toolbar application. You should submit images as JPG or PNG files. You can also submit all images in a single PDF file. Make sure to provide a brief explanation on the design choices you are making. Moreover, link your design to the abovementioned constraints.

Reflection (40%)

The reflection should be written in a file called `reflect.txt` and attached to the same blackboard submission as the in-lab image of the interface. Write explanations for the following design decisions you made. Do not give trivial answers but explore the topics in depth.

1. Explain how your interface prevents the user from making mistakes.
2. Explain why you designed the operations the way they are and how this design follows the design constraints.